Jinha Choi

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Research Interests

My research interest centers on robot learning for complex embodied systems, with a particular focus on humanoid locomotion, whole-body control, and adaptive behavior in complex environments. I am interested in developing scalable and generalizable learning methods that enable robots to navigate, balance, manipulate, and interact robustly in the real world. More broadly, my goal is to build principled robot learning systems that can transfer across tasks, environments, and embodiments, allowing humanoid robots to acquire useful behaviors from limited data and generalize beyond narrow training settings.

Education

Yonsei University, Underwood International College *Mar 2021 – Present*

Double Major in Computer Science
B.S. Candidate in Nano Science and Engineering (NSE)

GPA: 4.1 / 4.5

West Vancouver Secondary School, Canada *Sep 2014 – Jun 2020*

International Baccalaureate Diploma

Publications

Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport

Jinha Choi*, Jaehun Shon*, Jongwook Jeon, Jongmin Lee

Co-1st-author, 2026 NeurIPS submission

When AI Co-Scientists Fail: SPOT—a Benchmark for Automated Verification of Scientific Research

Guijin Son, Jiwoo Hong, Honglu Fan, Heejeong Nam, Hyunwoo Ko, Seungwon Lim, Jinyeop Song, **Jinha Choi**, Gonçalo Paulo, Youngjae Yu, Stella Biderman

Contributing co-author, arXiv

Research

KAIST HuGeLab Intern *June 2026 – Current*

Advised by Prof. Beomjoon Kim

Current research focuses on humanoid robot learning, with an emphasis on locomotion, whole-body control, and robust behavior in complex environments. I am working on learning-based approaches that enable humanoid robots to move through diverse settings, adapt to changing physical conditions, and coordinate balance, navigation, and interaction. More broadly, I am interested in scalable and generalizable methods that allow humanoid robots to acquire useful behaviors beyond narrow task-specific settings.

Yonsei DILLAB Intern *Jan 2026 – June 2026*

Advised by Prof. Jongmin Lee

Developed efficient multimodal flow policies via value-weighted optimal transport for offline reinforcement learning. Learned value-aware multi-step reference policies from offline datasets and distilled them into one-step target policies through a structured sample allocation formulation. Used entropic optimal transport over action samples from reference and target policies to improve multimodal policy distillation, achieving strong performance across diverse benchmark tasks.

Yonsei RLLAB Intern *Jul 2025 – Dec 2025*

Advised by Prof. Youngwoon Lee

Built and analyzed baseline pipelines for humanoid loco-manipulation in simulation, including motion-prior training from retargeted human-motion datasets. Authored an undergrad term paper assessing retargeting methodology and its effects on learned motion quality and stability. Studied reusable motion components as a path toward better transfer across tasks, with an emphasis on minimizing task-specific reward engineering.

Yonsei MIRLAB Intern *Sep 2024 – Dec 2024*

Advised by Prof. Youngjae Yu

Trained in lab workflows and research practices for robot learning. Studied multi-modal learning and built familiarity with the Isaac Sim simulation for RL environments. Contributed to an LLM benchmarking study by processing and labeling datasets and organizing classification schemes for experiments evaluating LLMs' ability to assess scientific papers.

Experience

Humanoid Fixed-Base Manipulation *Mar 2026 – June 2026*

Advised by Prof. Youngwoon Lee

Investigating exploration limitations of standard PPO for humanoid fixed-base manipulation in massively parallel simulation. Studying PPO-based variants such as SAPG and EPO, which maintain heterogeneous learner policies and aggregate their updates into a leader policy through importance-weighted optimization. Evaluating whether these approaches outperform standard PPO on humanoid manipulation benchmarks and exploring directions for improving exploration and policy learning.

A1-Challenge Autonomous Driving Competition *Mar 2026 – May 2026*

Qualified in Nov 2025; hosted by the Ministry of Trade and KIAPI

Developing an end-to-end autonomous driving policy in the MORAI simulator for competitive racing against university teams. Built a strong imitation-learning baseline using behavioral cloning and DAgger with targeted data collection on challenging turns, corner cases, and edge scenarios; currently exploring reinforcement learning in simulation to improve robustness and driving performance. Top-performing teams advance to real-vehicle autonomous driving on track.

Physical AI 26H Robotics Hackathon *Mar 2026*

Hosted by Kernel Academy and Doosan Robotics

Built a natural-language-guided pick-and-place system using Doosan E0509 robot arm, RGB-D sensing, vision-language reasoning, and ROS2 execution. The system processed spoken commands with Whisper, segmented scene objects with SAM2, identified pick and place targets through GPT-4o-based reasoning, localized end-effector coordinates from depth, and executed robot actions through calibrated camera-to-robot transformation and motion primitives. The full pipeline also included before/after verification and error-recovery logic for more reliable task execution.

Yonsei YAX *Mar 2026 – June 2026*

Advised by Prof. Shiho Kim

Member of Yonsei's robotics and autonomous driving club, participating in projects across both robotics and autonomous driving simulations. Engaged with imitation learning and reinforcement learning workflows in simulation, with emphasis on practical experience in robot learning and autonomous systems development.

SW · AI Business Project *Mar 2026 – June 2026*

Advised by Prof. Hyunsoo Kim

Developing a web-based travel-planning platform that recommends destinations, places to visit, and activities based on travel dates, trip context, and user preferences. The project aims to go beyond flight and hotel booking by helping users both select where to go and organize what to do during a given period. Planning and review features are designed to incorporate user travel logs and community-generated information through a personalized travel map.

OUTTA AI Bootcamp – Deep Learning Advanced *Jul 2024 – Aug 2024*

Summer bootcamp offered by Outta

Completed a two-month advanced deep learning bootcamp covering core architectures and generative models. In a five-member team, built an LLM-conditioned diffusion pipeline that converts text clothing descriptions into conditioning signals to generate virtual try-on images with consistent outfit changes.

Korea Defense Intelligence Command – Translator *Mar 2023 – Sep 2024*

Military Service

Completed military service as a translator/interpreter supporting intelligence operations. Translated and interpreted classified documents and meetings for KDIC and partner agencies, including the National Intelligence Service (NIS), U.S.-based counterparts National Geospatial-Intelligence Agency (NGA), and several counterparts of other allies.

Vex Robotics *Sep 2017 – Jun 2020*

Team lead of 1010B at WVSS Robotics

Competed in VEX Robotics Competition seasons Starstruck, In the Zone, and Turning Point. Led Team 1010B (12 team members) through the full robot-development cycle: game strategy analysis, mechanical/electrical design and build (wiring, assembly), programming motor/sensor control and autonomous control periods, and competitive driving. Coordinated with alliance teams during the world championship.

Awards and Honors

2nd Place, Physical AI 26H Robotics Hackathon *Mar 2026*

Won 2nd place in the Physical AI 26H Robotics Hackathon centered on natural-language-guided manipulation. Contributed to Isaac Sim environment setup, camera configuration, domain randomization, training and deployment support, and ROS2 transfer workflows for a single-arm manipulation pipeline.

Peer Mentor, Yonsei–Nexon RC Creative Platform *Mar 2024 – Dec 2024*

Selected as a peer mentor based on prior competition performance (former 1st-place team). Provided guidance to participating teams and received an official mentoring certificate.

Korea Defense Intelligence Command AI Security Competition *Jul 2024*

Awarded by Commander Sangho Moon

Received a top award in an AI security competition. Proposed an on-device AI framework for military systems with partial internet exposure, designed to support operational convenience while strengthening protection against security vulnerabilities and malicious misuse.

Honor Roll, Yonsei University *2022*

1st Place, Yonsei–Nexon RC Creative Platform *Mar 2021 – Dec 2021*

Won 1st place in a university-wide social-impact startup ideation competition co-hosted with Nexon. Led a 5-member team to design an automated revolving-door assistance system to improve safe building access for people with mobility impairments. Developed a 3D-printed prototype and presented the design rationale (safety, practicality, community impact) to judges and organizers; ranked 1st of 90 teams and awarded $10K.

Finalist, VEX Robotics World Championship — Kentucky, USA *2019*

Competed at the VEX Robotics World Championship (600 qualified teams from the world). Ranked as a Science Division finalist (top finish within a 100-team division), advancing to the division final and narrowly missing qualification to the round-robin finals against the top teams from other divisions.

1st Place, VEX Robotics Canada BC Provincial Championship *2019*

Led Team 1010B to qualify for Provincials via a regional championship, then won 1st place in British Columbia Provincials, earning qualification to the VEX Robotics World Championship.